MILESTONE 1

Project Objective

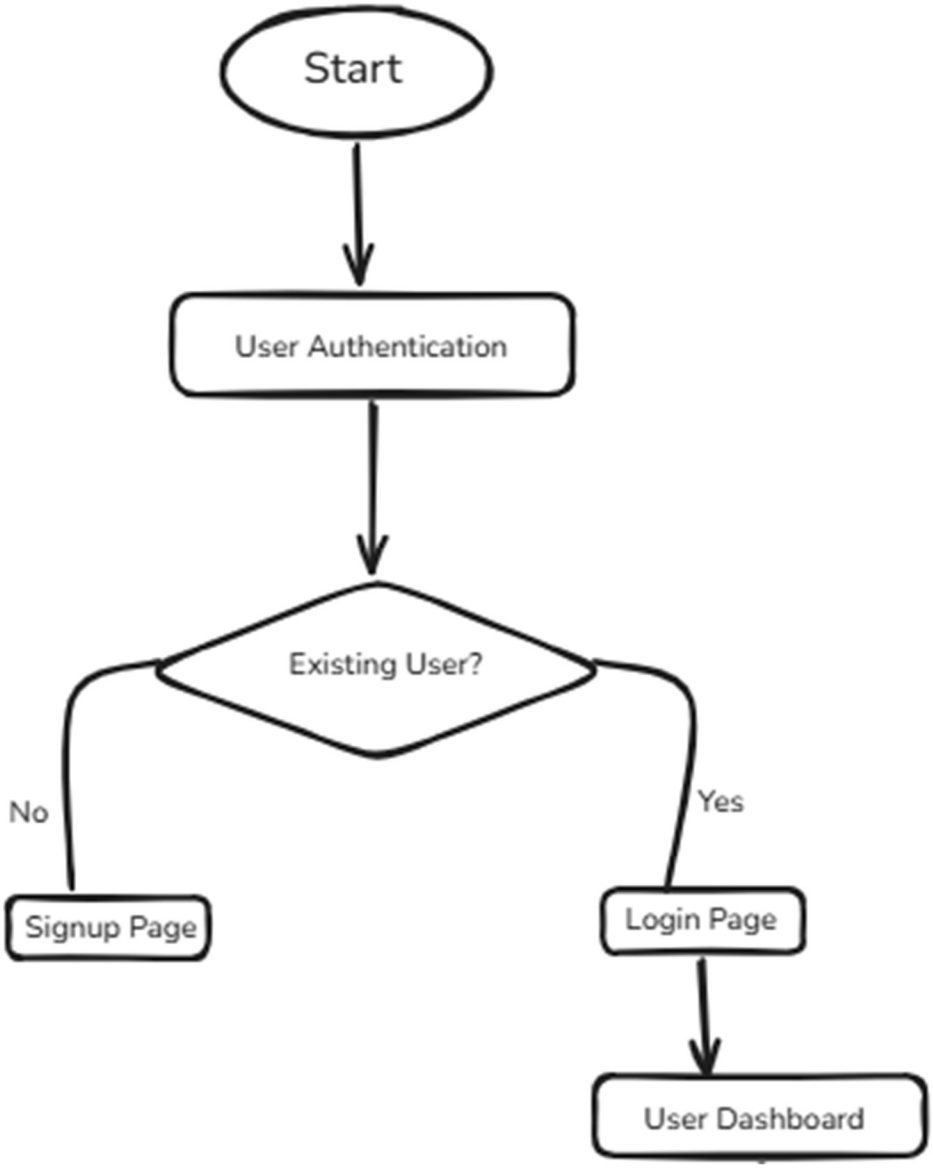
The primary objective of this project is to design and develop an intelligent, AI-powered Career Recommendation System as part of the **Infosys Springboard Internship Program**. The system aims to leverage resume analytics and automation technologies to provide personalized and data-driven career guidance.

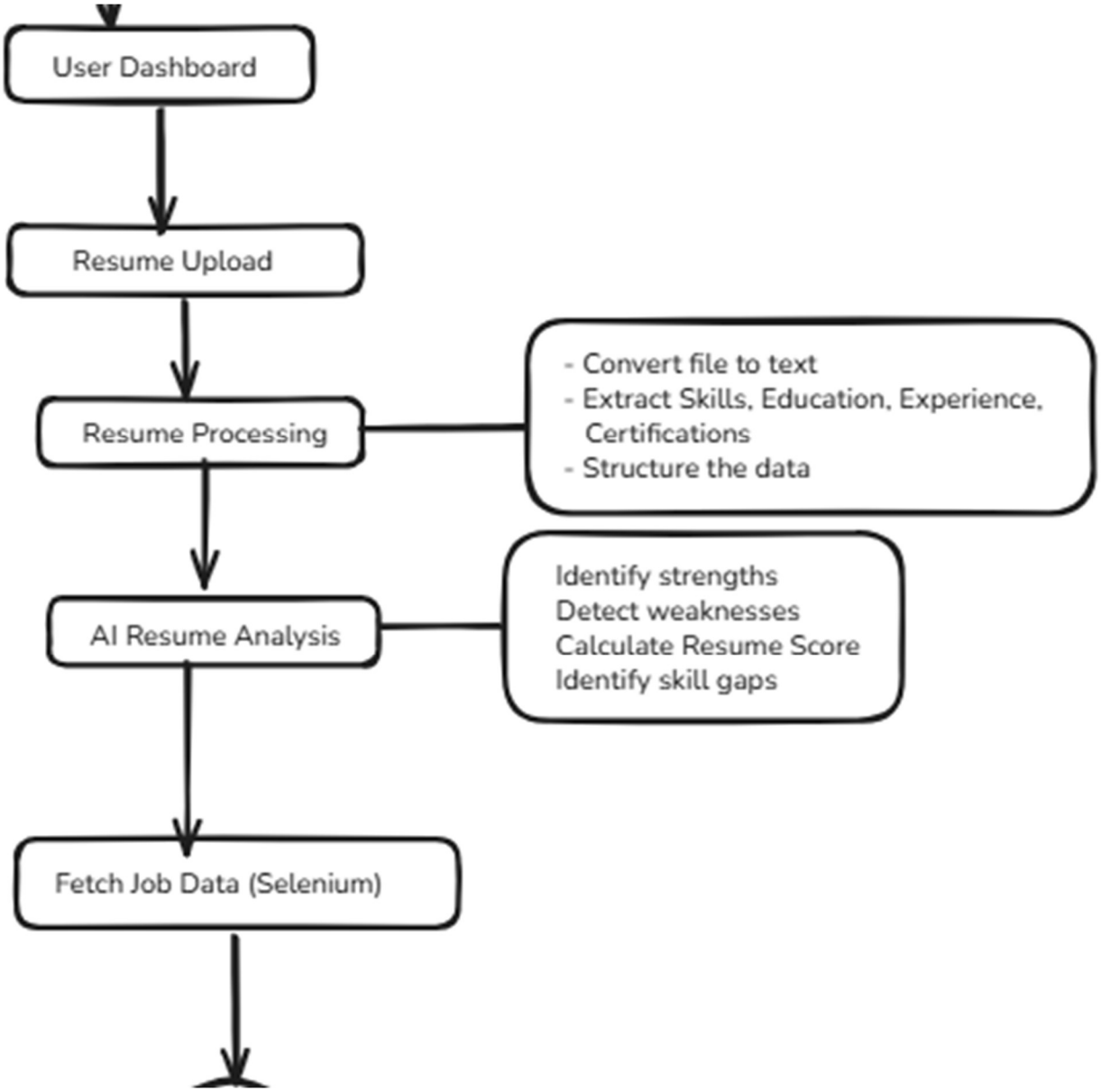
The key objectives of the proposed system are:

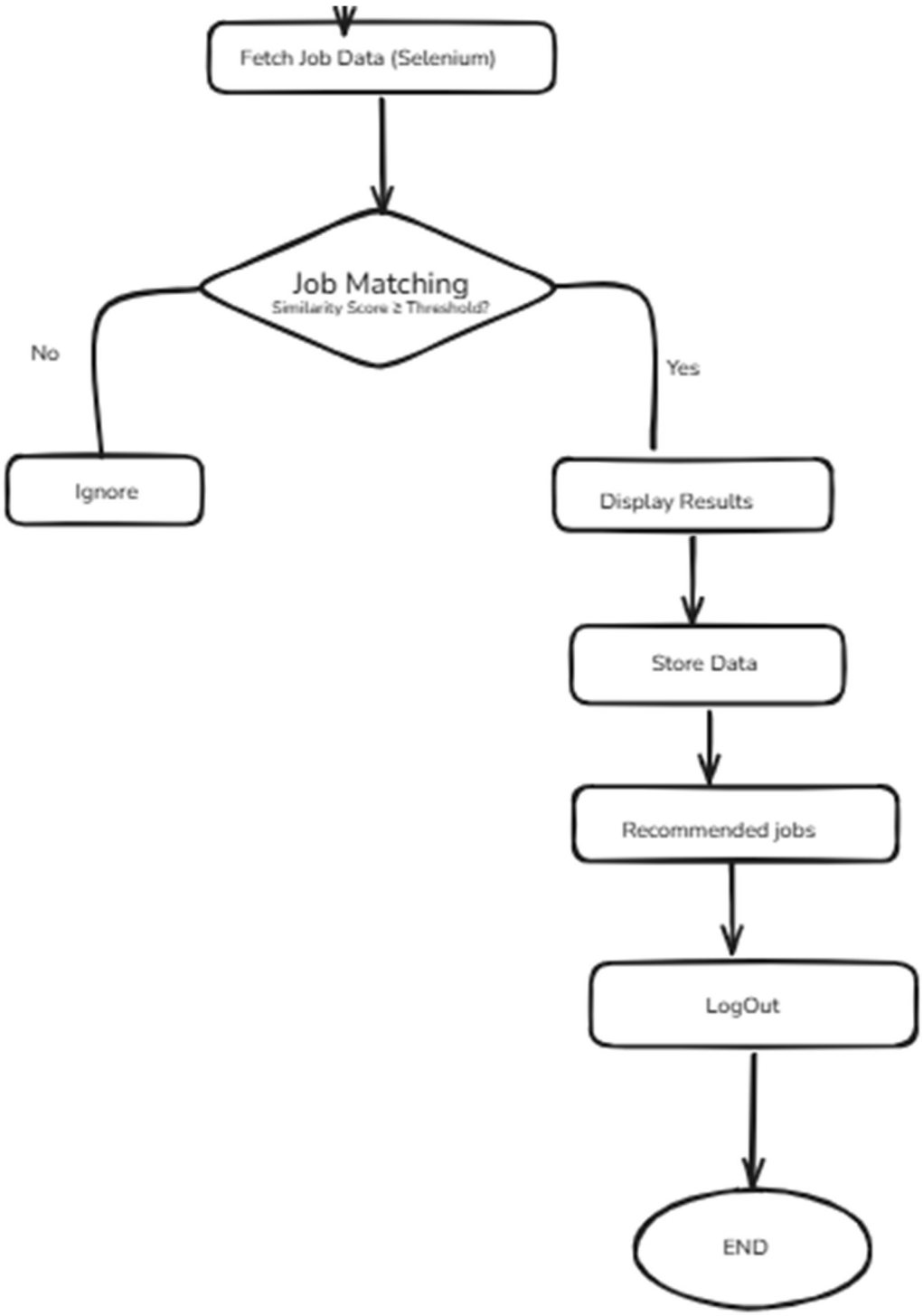
- To implement **resume analytics techniques** for building an intelligent career recommendation framework.
- To enable users to upload their resumes and automatically extract relevant information such as skills, education, experience, certifications, and achievements using advanced parsing techniques.
- To apply **Artificial Intelligence (AI) and Machine Learning (ML)** algorithms to analyze resumes and identify strengths, weaknesses, skill gaps, and experience levels.
- To utilize **Selenium automation** for collecting real-time employment-related data from online job portals.
- To generate accurate and personalized job recommendations by matching extracted resume data with job requirements.
- To develop an intuitive and user-friendly web interface using **Streamlit** for seamless user interaction.
- To implement a secure database management system for efficient storage, retrieval, and management of user data.

FlowChart

(https://excalidraw.com/#json=zaEBWoTlILrA_wwA_tZw8,2JjBJOgmp8AKNfi2b-v-Jg)




User Dashboard
Resume Upload
Resume Processing
- Convert file to text
- Extract Skills, Education, Experience, Certifications
- Structure the data
AI Resume Analysis
Identify strengths
Detect weaknesses
Calculate Resume Score
Identify skill gaps
Fetch Job Data (Selenium)


Fetch Job Data (Selenium)
Job Matching
Similarity Score ≥ Threshold?
No
Yes
Ignore
Display Results
Store Data
Recommended jobs
LogOut
END

Pseudo Code:

```
START

1. Initialize System
   - Connect to Database
   - Load AI/ML Model
   - Initialize Streamlit Interface
   - Initialize Selenium WebDriver

2. Display Login / Signup Page

3. IF user selects Signup THEN
       INPUT name, email, password
       IF email not already registered THEN
           Store user details in database
           Display "Signup Successful"
       ELSE
           Display "User Already Exists"
       ENDIF
       Redirect to Login
   ENDIF
```

```
4. IF user selects Login THEN
      INPUT email, password
      Validate credentials from database
      IF valid THEN
          Go to Dashboard
      ELSE
          Display "Invalid Credentials"
          Return to Login Page
      ENDIF
  ENDIF

5. Dashboard
  Display options:
      - Upload Resume
      - View Previous Reports
      - Logout

6. IF user selects Upload Resume THEN
      INPUT resume_file (PDF/DOCX)

      // Resume Processing
      Convert resume to text
```

Extract:

- Skills
- Education
- Experience
- Certifications
- Projects

Store extracted data in structured format

// Resume Analysis

Analyze strengths

Identify weaknesses

Detect skill gaps

Calculate resume_score

7. Fetch Job Data using Selenium

Open job portal

Search jobs using extracted skills

Scrape:

- Job Title
- Required Skills
- Experience Required

- Location

Store job data in database

8. Match Resume with Jobs

```
FOR each job in job_database DO
    similarity_score = compare(resume_skills, job_required_skills)

    IF similarity_score ≥ threshold THEN
        Add job to recommended_list
    ENDIF
END FOR
```

9. Rank Jobs

Sort recommended_list by similarity_score (Descending)

10. Display Results

Show:

- Resume Score
- Strengths
- Weaknesses
- Skill Gap Suggestions
- Top Recommended Jobs

11. Store analysis results in database

12. IF user selects Logout THEN

    End session

  ENDIF

END